

Is a structural-MRI foundation model worth its weight?

A four-rung classical morphometry ladder for the Dallas Lifespan Brain Study sMRI-FM benchmark

Morgan Hough
morgan.hough@gmail.com

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Abstract

The release of the MedARC `smri-fm` dataset, the publication of BrainIAC [AIM-KannLab contributors, 2026], and the MICCAI 2025 FOMO25 and SSL3D challenges won by Gordaliza et al. [2026] have renewed interest in structural-MRI foundation models (sMRI-FMs) as a general-purpose backbone for neuroimaging. Whether FMs provide a meaningful advance over classical morphometric features for core tasks — above all, brain-age prediction — has not been settled. In fact Schulz et al. [2025] show that for *disease detection*, simpler models beat complex ones despite losing on age MAE: ridge regression with 1 400 parameters produces larger patient-vs-control effect sizes than a 46M-parameter CNN on the same data. We therefore construct a **four-rung classical morphometry ladder** on the Dallas Lifespan Brain Study (DLBS, Park et al., 2025) and benchmark three sMRI foundation-model checkpoints against it under identical bias-correction regimes. This manuscript is generated with Pweave so every number and figure below updates as the experiment parquets land.

Current snapshot: 23 subjects, 60 sessions processed by FastSurfer VINN; 23 subjects, 60 sessions by T1Prep (CAT-Surface).

1 Introduction

Brain-predicted age, introduced by Franke et al. [2010] and popularised by Cole and colleagues [Cole et al., 2017], has become the single most-reported biomarker derived from structural MRI. A decade of architectural progress — from relevance-vector regression on VBM features, through Gaussian-process regression [Cole et al., 2017, Cole] to 3D CNNs [Peng et al., 2021] — has pushed UK Biobank mean absolute error (MAE) from ~ 5 years to ~ 2 years [Peng et al., 2021]. At the same time, multiple authors have shown that classical ridge- or kernel-based regressors on FreeSurfer or CAT12 morphometric features match deep networks within noise [More et al., 2023, Lombardi et al., 2021, Bashyam et al., 2020]. Gijssen et al. [2024] established in a 46 381-subject UK Biobank study that ResNet-50 and SwinT reach an indistinguishable 2.66 years MAE, with SwinT scaling better only above $\sim 25k$ training subjects. The apparent ceiling is set by label noise and site effects, not by model capacity.

Schulz et al. [2025] take this further and show that for disease detection, simpler models actually *beat* complex ones on patient-vs-control effect size even when they lose on age MAE. This finding motivates our evaluation plan: disease-effect-size is reported alongside age MAE for every arm.

2 Methods (pre-registration)

2.1 Data

We use the Dallas Lifespan Brain Study [Park et al., 2025], OpenNeuro `ds004856`, with 464 / 338 / 224 subjects across waves 1/2/3 (~ 1000 T1 sessions, Philips 3T Achieva, ~ 4 –5 year intervals, ages 20 to 90 at wave 1). External generalisation is evaluated on `ds003592` [Spreng et al., 2022]. Both were downloaded from OpenNeuro S3 on 2026-04-22.

2.2 Four classical morphometry rungs

Every pipeline runs inside a pinned Docker image from the `ghcr.io/m9h/neurocontainers-arm` registry for reproducibility on NVIDIA Grace-Blackwell arm64 hosts.

Rung 1 — SynthSeg volumes (MedARC baseline). The MedARC `preprocessing/pipeline.py` [MedARC-AI contributors, 2026]: `SynthStrip`, ANTs rigid to MNI152NLin2009cAsym, `mri_synthseg --parc --robust`. Aggregated to 71 ICV-normalised features per Bethlehem et al. [2022]. Implemented upstream by Ashtari [2026].

Rung 2 — SynthSeg + N4. Tests whether an upstream bias step helps a bias-robust segmentation network.

Rung 3 — FastSurfer VINN. `aparc.DKTatlas+aseg.deep VINN + CerebNet + HypVINN`. Isolates seg-network quality from bias handling. Runs in `ghcr.io/m9h/fastsurfer-arm:6b6b985-arm.1`.

Rung 4 — T1Prep thickness + Jacobian. T1Prep v0.3.3 [Gaser et al., 2026] (DeepMriPrep + AMAP + CAT-Surface). Adds cortical thickness per Desikan–Killiany ROI and nonlinear-warp log-Jacobian — the two feature axes no volumetric seg captures. Runs in `ghcr.io/m9h/t1prep-arm:v0`.

2.3 Foundation-model arms

Arm	Design	Reference
BrainIAC	SimCLR ViT, 32k pretraining scans	[AIM-KannLab contributors, 2026]
FOMO25 mmunetvae	3D U-Net MAE, 11 187 FOMO-60K scans	[Gordaliza et al., 2026]
AnatCL	anatomy-supervised contrastive, OpenBHB 3 984	[Barbano et al., 2024]
MedARC smri-fm	TBD; reported when release lands	[MedARC-AI contributors, 2026]

2.4 Bias correction

For every arm we report MAE under four regimes: raw, Cole-Smith [Smith et al., 2019], Beheshti [Liang et al., 2019], and Zhang age-level [Zhang et al., 2023].

2.5 Disease-effect-size reporting per Schulz et al. [2025]

For each arm we report patient-vs-control effect size (Cohen’s d) on an AD-spectrum labelling derived from DLBS participant metadata.

2.6 Longitudinal stability per Vidal-Piñeiro et al. [2021]

For each returning subject we compute $\Delta\text{BAG}_{i,t}$ between waves and test (a) within-subject stability (ICC), and (b) correlation with change in morphometric summaries. DLBS’s three-wave structure is the first public venue where this test can be run on an sMRI-FM.

3 Current results

3.1 Dataset coverage

Tool	Subjects	Sessions	Feature rows
FastSurfer VINN	23	60	10020
T1Prep	23	60	8114

3.2 FastSurfer–T1Prep concordance on sub-1003

Figure 1: DK ROI volume (FastSurfer VINN) vs cortical thickness (T1Prep CAT-Surface) for DLBS sub-1003 across three waves.

3.3 Ridge brain-age baseline (current parquet)

Arm & correction	n scans	MAE (yr)	RMSE	R^2	Pearson r
SynthSeg vols, TIV-norm (MedARC default) / raw	60	5.99	7.54	0.86	0.93
SynthSeg vols, TIV-norm (MedARC default) / cole	60	6.20	7.68	0.85	0.93
SynthSeg vols, TIV-norm (MedARC default) / beheshti	60	4.84	6.00	0.91	0.96
SynthSeg vols, TIV-norm (MedARC default) / zhang	60	4.71	6.28	0.90	0.95
SynthSeg vols, raw (MedARC pipeline) / raw	60	7.15	8.59	0.81	0.90
SynthSeg vols, raw (MedARC pipeline) / cole	60	7.58	9.51	0.77	0.90
SynthSeg vols, raw (MedARC pipeline) / beheshti	60	6.04	7.59	0.85	0.93
SynthSeg vols, raw (MedARC pipeline) / zhang	60	5.85	7.30	0.86	0.93
FS aseg+DKT volumes, TIV-norm (rung 3) / raw	60	7.53	9.50	0.77	0.88
FS aseg+DKT volumes, TIV-norm (rung 3) / cole	60	8.34	10.68	0.71	0.88
FS aseg+DKT volumes, TIV-norm (rung 3) / beheshti	60	6.79	8.70	0.81	0.92
FS aseg+DKT volumes, TIV-norm (rung 3) / zhang	60	5.48	7.76	0.85	0.93
FS aseg+DKT volumes, raw / raw	60	7.48	9.21	0.78	0.89
FS aseg+DKT volumes, raw / cole	60	8.73	10.33	0.73	0.89
FS aseg+DKT volumes, raw / beheshti	60	7.25	8.57	0.81	0.92
FS aseg+DKT volumes, raw / zhang	60	5.85	7.70	0.85	0.93
SynthSeg + T1Prep-thickness concat / raw	60	7.21	8.50	0.82	0.91
SynthSeg + T1Prep-thickness concat / cole	60	7.72	9.27	0.78	0.91
SynthSeg + T1Prep-thickness concat / beheshti	60	6.77	8.13	0.83	0.93
SynthSeg + T1Prep-thickness concat / zhang	60	6.75	7.98	0.84	0.92
FS + T1Prep-thickness concat / raw	60	6.65	8.17	0.83	0.91
FS + T1Prep-thickness concat / cole	60	7.38	8.96	0.80	0.91
FS + T1Prep-thickness concat / beheshti	60	6.20	7.51	0.86	0.94
FS + T1Prep-thickness concat / zhang	60	6.31	7.40	0.86	0.93
T1Prep tissue (CSF/GM/WM), TIV-norm / raw	60	7.42	9.24	0.78	0.89
T1Prep tissue (CSF/GM/WM), TIV-norm / cole	60	8.28	10.29	0.73	0.89
T1Prep tissue (CSF/GM/WM), TIV-norm / beheshti	60	6.98	8.67	0.81	0.92
T1Prep tissue (CSF/GM/WM), TIV-norm / zhang	60	6.58	8.22	0.83	0.92
T1Prep thickness (rung 4) / raw	60	8.03	9.53	0.77	0.88
T1Prep thickness (rung 4) / cole	60	8.97	10.57	0.72	0.88
T1Prep thickness (rung 4) / beheshti	60	7.55	8.90	0.80	0.91
T1Prep thickness (rung 4) / zhang	60	7.20	8.87	0.80	0.91
T1Prep thickness+area concat / raw	60	10.17	12.21	0.62	0.81
T1Prep thickness+area concat / cole	60	11.82	14.48	0.47	0.81
T1Prep thickness+area concat / beheshti	60	9.24	11.31	0.68	0.87
T1Prep thickness+area concat / zhang	60	9.04	10.89	0.70	0.87
T1Prep area alone / raw	60	16.32	19.00	0.08	0.46
T1Prep area alone / cole	60	31.60	38.26	-2.72	0.46
T1Prep area alone / beheshti	60	11.46	13.87	0.51	0.82
T1Prep area alone / zhang	60	10.34	12.47	0.61	0.84
FS + T1Prep-all concat / raw	60	7.57	9.05	0.79	0.89
FS + T1Prep-all concat / cole	60	8.28	10.12	0.74	0.89
FS + T1Prep-all concat / beheshti	60	6.77	8.27	0.83	0.92
FS + T1Prep-all concat / zhang	60	6.80	8.15	0.83	0.92
BrainIAC 768-d (FM) / raw	60	13.73	16.20	0.33	0.60
BrainIAC 768-d (FM) / cole	60	21.26	26.13	-0.73	0.60
BrainIAC 768-d (FM) / beheshti	60	10.08	12.39	0.61	0.85
BrainIAC 768-d (FM) / zhang	60	9.61	11.71	0.65	0.86
FS + T1Prep + BrainIAC concat / raw	60	8.31	9.56	0.77	0.88
FS + T1Prep + BrainIAC concat / cole	60	8.82	10.64	0.71	0.88
FS + T1Prep + BrainIAC concat / beheshti	60	7.47	9.00	0.79	0.91
FS + T1Prep + BrainIAC concat / zhang	60	7.35	8.87	0.80	0.91

4 Infrastructure contributions

All pipelines run on an NVIDIA DGX Spark (GB10 Grace-Blackwell, arm64). Upstream neuroimaging containers are almost entirely amd64 and fall back to QEMU emulation on Grace, which is unusably slow for TF/PyTorch inference. To close this gap we contribute:

- `ghcr.io/m9h/t1prep-arm:v0.3.3-arm.1` — arm64 T1Prep with three Blackwell-specific patches pending upstream PR (deterministic-algorithm warn_only fallback, weight-cache write permissions, case-sensitive Neuromorphometrics atlas filenames).
- `ghcr.io/m9h/fastsurfer-arm:6b6b985-arm.1` — arm64 FastSurfer seg_only, NGC PyTorch 24.12 base. Published in response to Deep-MI/FastSurfer issue #716.
- `ghcr.io/m9h/medarc-arm` (pending build) — MedARC’s `smri-fm/preprocessing/pipeline.py` on arm64 via locally built FreeSurfer 8.2.0 .debs.

Recipes + auto-build workflow at <https://github.com/m9h/neurocontainers-arm>.

5 Pre-registered hypotheses

1. *Cross-sectional ceiling*. FM + ridge and rung-4 (T1Prep thickness + Jacobian) + ridge match within 0.3 years MAE on wave-1 held-out subjects.
2. *Bias persistence*. Regression-dilution slope α indistinguishable across preprocessing and FM arms.
3. *Longitudinal disconnect*. Within-subject Δ BAG correlates weakly ($|r| < 0.2$) with within-subject morphometric change, per Vidal-Piñero et al. [2021], for every arm.
4. *Disease-effect-size inversion*. Per Schulz et al. [2025], rungs 1–4 have higher patient-vs-control effect size than any FM + ridge despite losing on age MAE.

6 Discussion (placeholder)

To be written after ridge numbers land. The contribution is not a new model but an apples-to-apples evaluation that pins down where the sMRI-FM proposition is and is not load-bearing.

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